

5. EEPROM stands for

- A. Erasable Edit Programmable Read Only memory
- B. Erasable Edge Programmable Read Only memory
- C. Electrically Erasable Programmable Read Only memory
- D. Electrically Edit Programmable Read Only memory

4. The ARM7TDMI

The ARM7TDMI is a member of the Advanced RISC Machines (ARM) family of general purpose 32-bit microprocessors, which offer high performance for very low power consumption and price.

Advanced RISC machine (ARM) is the first reduced instruction set computer (RISC) processor for commercial use, which is currently being developed by ARM Holdings

- ▶ **Advanced RISC machine (ARM) is the reduced instruction set computer (RISC) processor, which is created by ARM Holdings**
- ▶ **90% of the mobile markets ruled by the ARM architecture technology**

Applications

Embedded Solutions

- Smart Cards • Automotive • Dashboard Controls

Enterprise Solutions

- Flash Cards • Hard Disks • Printers

Home Solutions

- Still Cameras • Digital TV • Set Top Box

Mobile Solutions

- Bluetooth • PDA • Smart Phone

Emerging Applications

- Gaming • Robot • And much more....

ARM 7 ARCHITECTURE Features

- 32-bit RISC-processor core (32-bit instructions)
- 37 pieces of 32-bit integer registers (16 available)
- Thumb instruction set
- Pipelined (ARM7: 3 stages)
- Cached (depending on the implementation)
- Von Neuman-type bus structure (ARM7), Harvard (ARM9)
- Debug Interface
- Embedded ICE macrocell
- Jazelle DBX(Direct Bytecode eXecution)
- 8 / 16 / 32 -bit data types
- 7 modes of operation (usr, fiq, irq, svc, abt, sys, und)

32-bit RISC-processor core

RISC Means Reduced Instruction Set Computer

THUMB set

- ▶ The THUMB set's 16-bit instruction length allows it to approach twice the density of standard ARM code while retaining most of the ARM's performance advantage over traditional 16-bit processor using 16-bit registers.

This is possible because THUMB code operates on the same 32-bit register set as ARM code

- ▶ The key idea behind THUMB is that of a super-reduced instruction set.
- ▶ Essentially ARM7TDMI processor has two instruction sets:
 - ▶ the standard 32-bit ARM set
 - ▶ a 16-bit THUMB set

pipelining

- ▶ It is used speed up the no. of execution per unit time for processor by instruction parallelism
- ▶ The process of fetching the next instruction when the present instruction is being executed is called as **pipelining**
- ▶ **Pipelining** is a technique that implements a form of parallelism called instruction-level parallelism within a single **processor**.
- ▶ It therefore allows faster CPU throughput (the number of instructions that can be executed in a unit of time) than would otherwise be possible at a given clock rate.

Instruction execution without pipeline

The execution of instruction performed one by one, after complete the execution of an instruction the next instruction is fetched from the memory

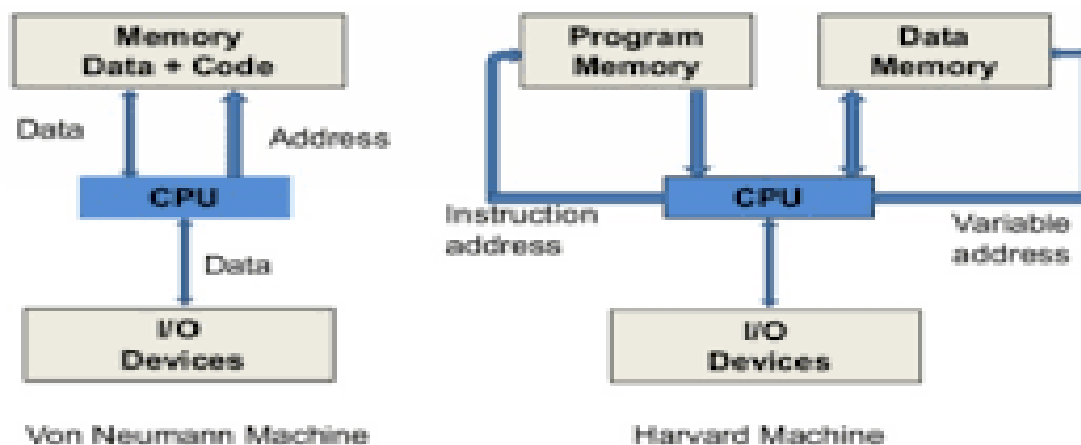
Suppose, these instructions are executed sequentially, and it takes the 1-unit clock cycle for each step to run. So, it would take 12-clock cycles(3×4) to execute these instructions.

Cache

- ▶ **Cache** is a small amount of memory which is a part of the CPU - closer to the CPU than RAM . It is **used** to temporarily hold instructions and data that the CPU is likely to reuse.

Harvard architecture vs Von-Neumann architecture

- ▶ In **Harvard architecture**, the CPU is connected with both the data memory (RAM) **and** program memory (ROM), separately.
- ▶ In **Von-Neumann architecture**, there is no separate data **and** program memory. Instead, a single memory connection is given to the CPU.



Debug Interface

- ▶ On chip unit for testing called as JTAG interface
- ▶ JTAG stands for Joint Test Action Group and defines the set of standards for testing the functionality of hardware.
- ▶ There is a set of scan cells located at the boundaries

Embedded ICE macrocell

- ▶ Macro cell ICE(In Circuit Emulator) is used to enable the testing .
- ▶ This unit is powered by breakpoint and watch point register and control & status register
- ▶ All the register work together to halt the ARM core to read status and thus do active debugging

Jazelle DBX(Direct Bytecode eXecution)

- ▶ ARM processors to execute Java byte code in hardware as a third execution state along with existing ARM and Thumb mode
- ▶ Useful to increase the execution speed of Java ME games and applications

Operating Modes

- ▶ ARM has seven operating modes:
 - ▶ User : Unprivileged mode under which most tasks run
 - ▶ FIQ(Fast Interrupt Request): Entered on a high priority interrupt request
 - ▶ IRQ(Interrupt Request): Entered on a low priority request
 - ▶ Supervisor: Entered on reset and when a software interrupt instruction is executed.
 - ▶ Abort: used to handle memory access violations
 - ▶ Undef: used to handle undefined instructions
 - ▶ System: privileged mode using the same register as user mode

4.3.Memory Interface

- ▶ The ARM processor has Von Neumann Architecture, with 32 bit data bus carrying both instruction and data
- ▶ Only the load, store, swap instructions can access from the memory

ARM Processor access the memory by four different types

1. Non sequential Bus Cycle

2. Sequential Bus Cycle

3. Internal Cycle

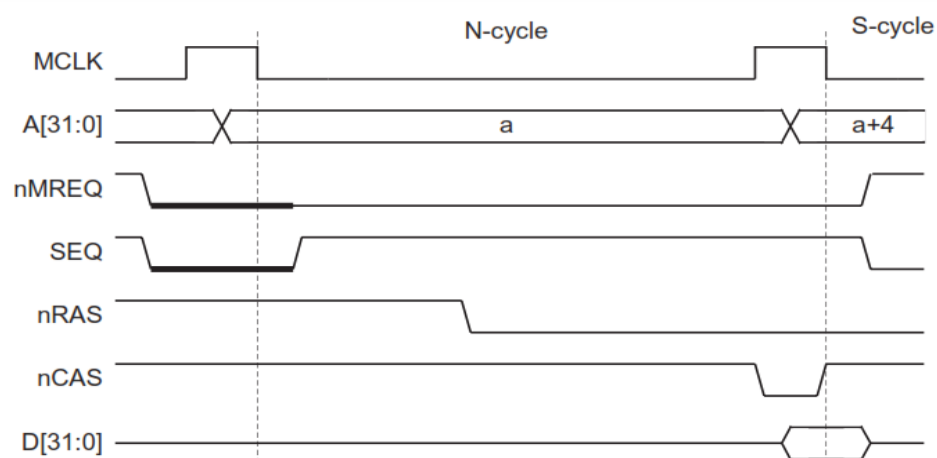
4. Co-processor Cycle

Processor determines the type transfer cycle takes place based on the two signals nMREQ and SEQ

nMREQ	SEQ	BUS Cycle	Description
0	0	N Cycle	Non sequential Bus Cycle
0	1	S- Cycle	Sequential Bus Cycle
1	0	I-Cycle	Internal Cycle
1	1	C-Cycle	Co-processor Cycle

1. Non sequential Cycle

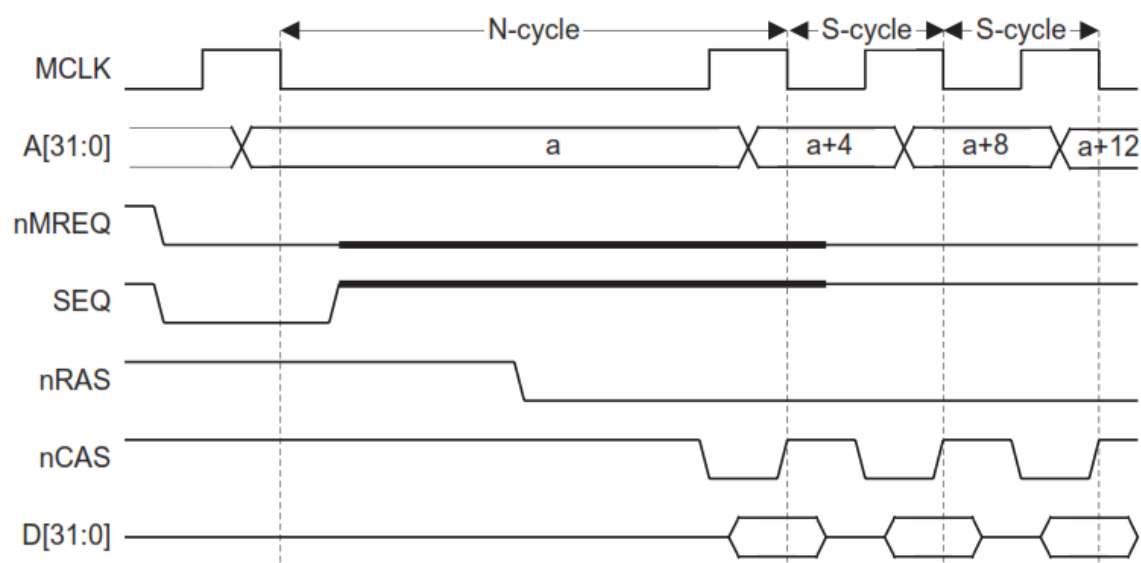
It is a type of memory access cycle in which access the memory through memory location is unrelated to the preceding cycle transfer.



Sequential Cycle

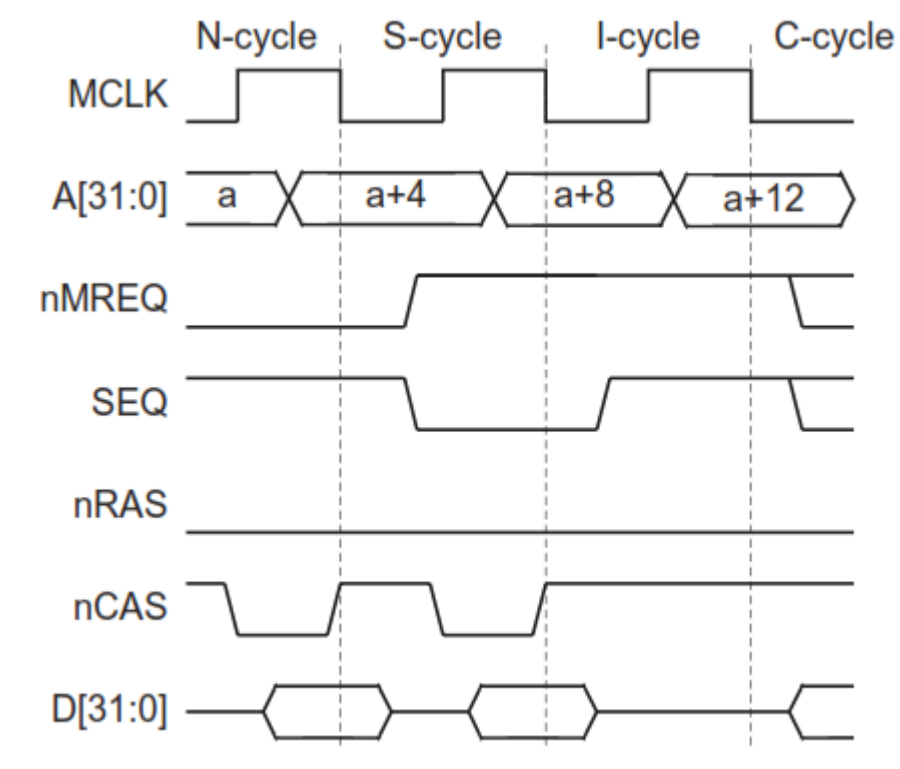
It is a type of memory access cycle in which access the memory through memory location is related to the preceding cycle transfer.

It is used generally program access from the memory and used for burst data transfer



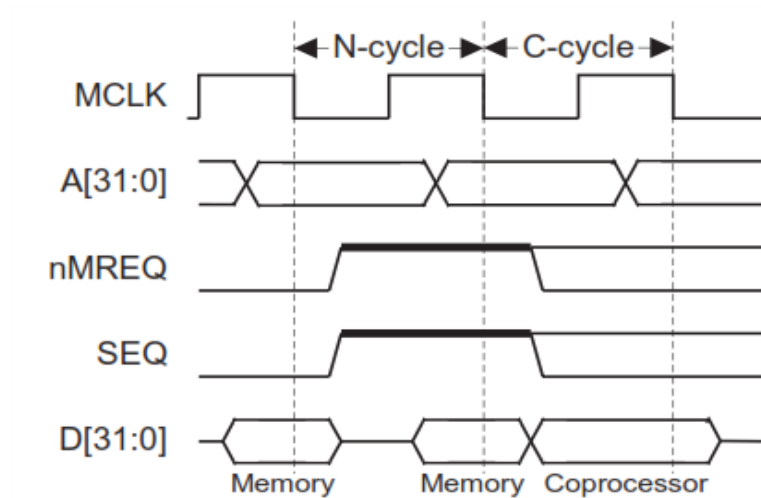
Internal Cycle

In this bus cycle no memory access performed only an internal function to be executed



Co-Processor register transfer Cycle

During this cycle, ARM7TDMI processor access the data from or to the Co-processor .memory access does not initiate the during the cycle.

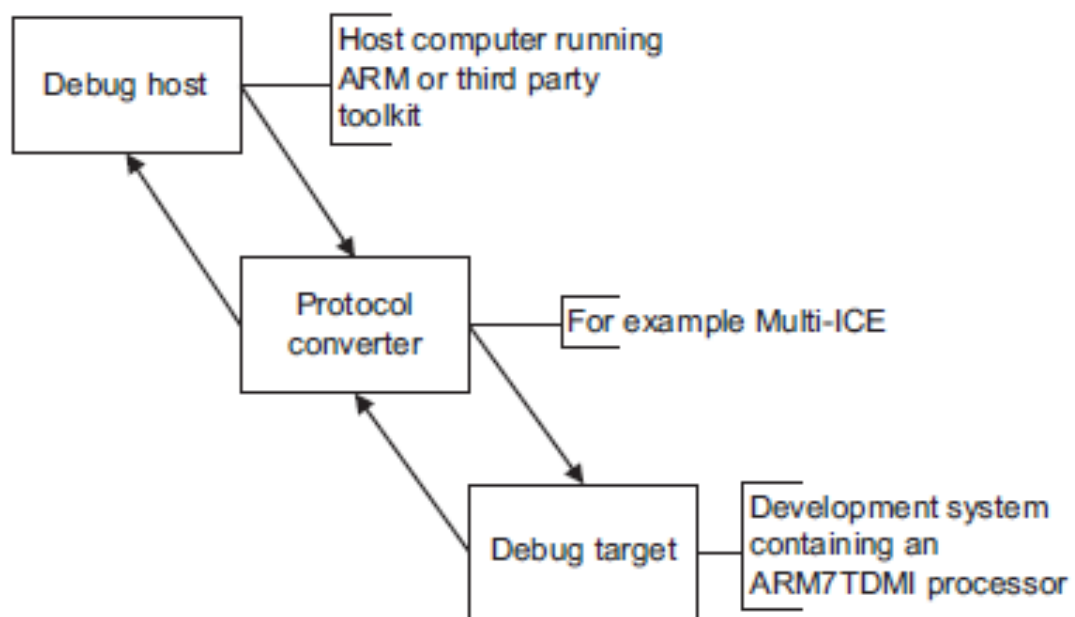


4.4.Debug Interface

- ▶ The debug interface is based on IEEE Std. 1149.1-1990, Standard Test Access Port and Boundary Scan Architecture.
- ▶ The ARM7TDMI processor contains hardware extensions for advanced debugging features
- ▶ These make it easier to develop application software, operating systems and hardware itself.

Stages of the Debug

- ▶ A request on one of the external debug signal, or on an internal functional unit known as EmbeddedICE Logic forces into debug state
- ▶ A break point, an instruction fetch
- ▶ A watch point, a data access
- ▶ An external debug request



Debug Host

- ▶ The Debug host is computer that is running a software debugger such as the Arm Debugger for Windows.

- ▶ The debug host allows to issue high level commands such as setting breakpoints of examining the contents of memory

Protocol Converter

- ▶ The protocol converter communicates with high commands issued by the debug host and the low level commands of JTAG interface.
- ▶ It interfaces to the host through an interface such as an enhanced parallel port.

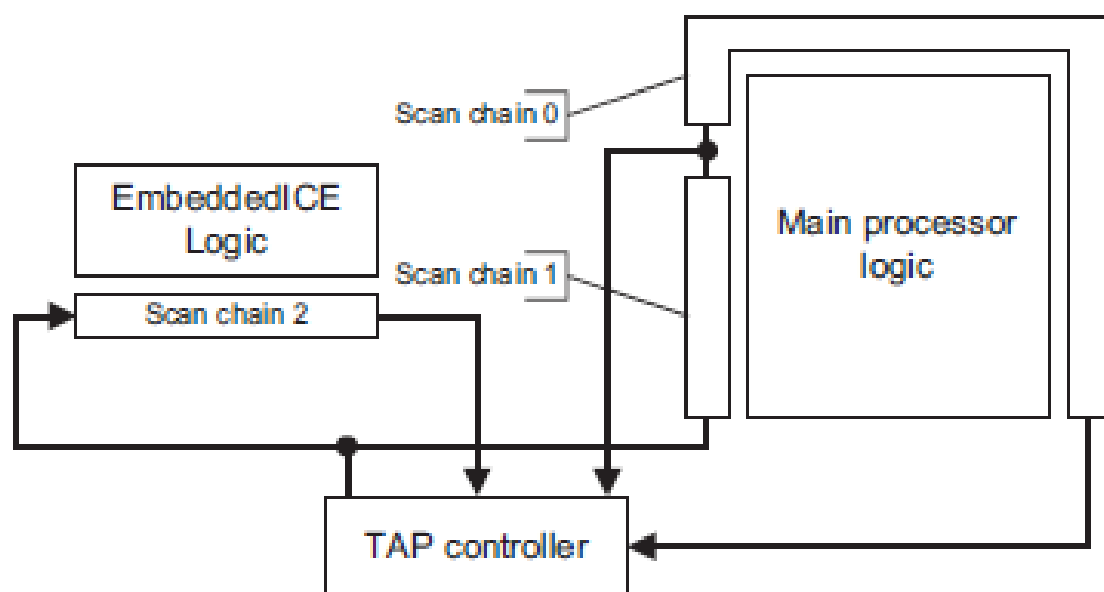
Debug Target

Debug target consists of EmbeddedICE Logic and TAP Controller

- ▶ The EmbeddedICE Logic

This is a set of registers and comparators used to generate debug exceptions such as breakpoints
- ▶ The TAP Controller

This controls the action of the scan chains using a JTAG serial interface



5.Post test-Multiple Choice Questions

The ARM core uses ____ Architecture.

- a) **RISC** b) CISC c) Both D) none

2. ARM Processor specifically designed for to reduce ____

a) Size b) Power Consumption **C) both** d) none.

3. ARM Processor core is a key component of ____ bit embedded system.

a) 8 b) 16 **c) 32** d) 64

4. ARM means ____

a) **Advance Risc Machine** b) Advance Review machine c) Advance Risc mechanism
d) All

5. RISC means ____

a) **Reduced Instruction set computer** b) Reduced Instruct set computer c) both d)
None

6. Conclusion

Hence the in this module we studied about the ARM Architecture in detail

7. References

1. ARM System Developer's Guide – Andrew N. Sloss.
2. ARM Architecture Reference Manual - David Seal.
3. ARM System-on-Chip Architecture (2nd Edition) by Steve Furber.

8. Assignment

1. Describe about the features of ARM7TDMI architecture
2. Write short notes about the Memory interface bus signal
3. Mention importance of the Debug interface bus signal